

TB Center

DETAILED ENGLISH USER MANUAL

A phase-conscious stereo centering processor with timing alignment, width control, lookahead analysis, and optional multiband weighting.



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Host-visible endpoints documented: 12

1. Product purpose

A phase-conscious stereo centering processor with timing alignment, width control, lookahead analysis, and optional multiband weighting.

Stereo recordings can lean unpredictably, wander as performers move, or feel unfocused because level and arrival-time differences disagree. Simple balance controls can move the image but cannot correct timing, and mono summing can remove valuable width. TB Center is intended for dialogue, stereo microphones, instruments, buses, and archival material that need a more stable center without indiscriminately collapsing the recording.

What result it creates

- A more stable dominant image
- Reduced left/right timing disagreement
- Controlled width after recentering
- Frequency-dependent correction when multiband mode is enabled

Signal flow

Input analysis -> direction and timing estimate -> optional four-band weighting -> center correction -> M/S width stage -> output

2. Complete feature guide

Intensity and Response

Intensity sets how strongly the detected offset is corrected. Response determines how quickly the detector follows movement: slower settings feel stable and natural, while faster settings track rapidly changing material.

Lookahead analysis

Lookahead gives the detector more future context and can make fast centering decisions cleaner. It adds processing latency and is best changed while playback is stopped when the host does not support dynamic latency changes smoothly.

Time Center

Time Center reduces inter-channel arrival-time differences. Use it when the image remains smeared after level centering; reduce it when the original stereo timing is an important part of the space.

M/S Width

The final mid/side stage ranges from mono-compatible narrowing through the original width to deliberate widening. Recheck phase correlation and mono compatibility when working above 100 percent.

Multiband weighting

Multiband enables independent Low, Mid-Low, Mid-High, and High correction weights. Set a band below 100 percent to preserve more of its original placement, or above its practical neutral influence only when the current build allows it.

Programs and visualization

Factory programs provide repeatable starting points for dialogue, natural centering, moving sources, bus recentering, and air-preserving width. The display is an analysis aid; make final decisions by ear and in mono.

3. Quick start

1. Insert the plug-in on a stereo track and begin with Init.
2. Set Intensity while listening for a stable center rather than maximum narrowness.
3. Adjust Response so movement is followed without audible image pumping.
4. Use Time Center only as far as needed to remove timing smear.
5. Set M/S Width for the final spatial size.
6. Enable Multiband only when different frequency regions need different correction strengths.

4. Practical workflows

Moving dialogue

1. Start with Natural Center or Locked Dialogue.
2. Raise Response until head movement is controlled.
3. Keep low-band correction firm for intelligibility.
4. Reduce high-band weighting if room reflections become unnaturally fixed.

Expected result: Speech remains centered and readable while the room retains useful dimension.

Stereo bus recentering

1. Start at moderate Intensity.
2. Leave Time Center below maximum unless timing is visibly unstable.
3. Use M/S Width near 100 percent, then trim only as required.
4. Check mono and compare equal loudness.

Expected result: The mix gains focus without losing its intended stereo scale.

5. Automation, gain staging, and session use

- Automate only controls that serve a clear musical transition. Fast movement of frequency, time, pitch, mode, oversampling, or algorithm selectors may intentionally create artifacts or require a host-safe transition.
- Match perceived loudness before judging quality. A louder setting will often appear better even when the processing decision is not better.
- Use the plug-in bypass endpoint for comparison and preserve an unprocessed source or earlier project version.
- Factory programs are starting points. Loading a program changes multiple parameters; save a new DAW preset after adapting it to the source.
- The parameter appendix is captured from the installed VST3 host contract. It lists every host-visible endpoint, its displayed range/options, default, automation status, and identifier.

6. Limits, cautions, and troubleshooting

- The processor cannot create missing stereo information.
- Excessive correction can make ambience unnaturally static.
- Widening can expose phase problems that were already present.

- No audible change: confirm the plug-in and relevant sub-block are enabled, Mix/Amount is not at a neutral value, and the source contains energy in the processed region.
- Unexpected level: return input, output, send, feedback, and parallel-mix controls to conservative values, then rebuild the setting one block at a time.
- Automation does not match the panel: reload the project, confirm the DAW is addressing the current plug-in version, and check whether a factory program or state recall replaced values.
- Clicks or transitions: avoid sample-instant changes to algorithm, time, pitch, mode, or oversampling selectors unless the sound is intentional.

7. Complete host parameter reference

This appendix contains all 12 parameters exposed by the current installed VST3 to a host. Repeated EQ-band endpoints are intentionally listed rather than collapsed. Discrete options are printed in full when the host contract exposes them.

Parameter	Range / options	Default	Host status	Function
Band High VST3 ID 282509655 Source ID bandhigh	0.0 to 100.0	100.0	Automatable	Host-automatable control for Band High. Its full range and default are shown in this table; use it with the related feature section above.
Band Low VST3 ID 1810232575 Source ID bandlow	0.0 to 100.0	100.0	Automatable	Host-automatable control for Band Low. Its full range and default are shown in this table; use it with the related feature section above.
Band Mid-High VST3 ID 396600341 Source ID bandmidhigh	0.0 to 100.0	100.0	Automatable	Host-automatable control for Band Mid-High. Its full range and default are shown in this table; use it with the related feature section above.
Band Mid-Low VST3 ID 1051902593 Source ID bandmidlow	0.0 to 100.0	100.0	Automatable	Host-automatable control for Band Mid-Low. Its full range and default are shown in this table; use it with the related feature section above.
Bypass VST3 ID 1652125811	Off, On	Off	Automatable; Bypass	Turns the complete plug-in processing path off or on through the host-visible bypass endpoint.
Intensity VST3 ID 499324979 Source ID intensity	0.0 to 200.0	100.0	Automatable	Sets how strongly the named process affects the signal.
Lookahead VST3 ID 1131625090 Source ID lookahead	0 to 100	20	Not automatable	Provides future-signal context to the associated dynamics or limiter stage and may add latency.
M/S Width VST3 ID 113126854 Source ID width	0.0 to 200.0	100.0	Automatable	Sets stereo spread or the lateral distribution of the processed signal.
Multiband VST3 ID 940938990 Source ID multiband	Off, On	Off	Automatable	Host-automatable control for Multiband. Its full range and default are shown in this table; use it with the related feature section above.
Program VST3 ID 1886548852	Init, Locked Dialogue, Natural Center, Fast Motion, Mix Recenter, Wide Keep Air	Init	Automatable	Selects a factory program. Loading a program recalls a coordinated set of plug-in parameters.
Response VST3 ID 1807160385 Source ID response	0.0 to 100.0	75.0	Automatable	Host-automatable control for Response. Its full range and default are shown in this table; use it with the related feature section above.
Time Center VST3 ID 38272088 Source ID timealign	0.0 to 100.0	100.0	Automatable	Host-automatable control for Time Center. Its full range and default are shown in this table; use it with the related feature section above.

Documentation basis

Prepared from the current public catalog, current local product brief and implementation notes where available, existing English manuals where available, and a direct scan of the installed VST3 host contract. Internal implementation details that are not user-facing are intentionally excluded; all user-facing features and host-visible controls are documented.